



FAKULTI TEKNOLOGI MAKLUMAT DAN KOMUNIKASI

DITU 3934

SYSTEM DEVELOPMENT WORKSHOP

RAKAN AKADEMIK FTMK

DESIGN

LECTURER'S NAME: ROSMIZA WAHIDA BINTI ABDULLAH

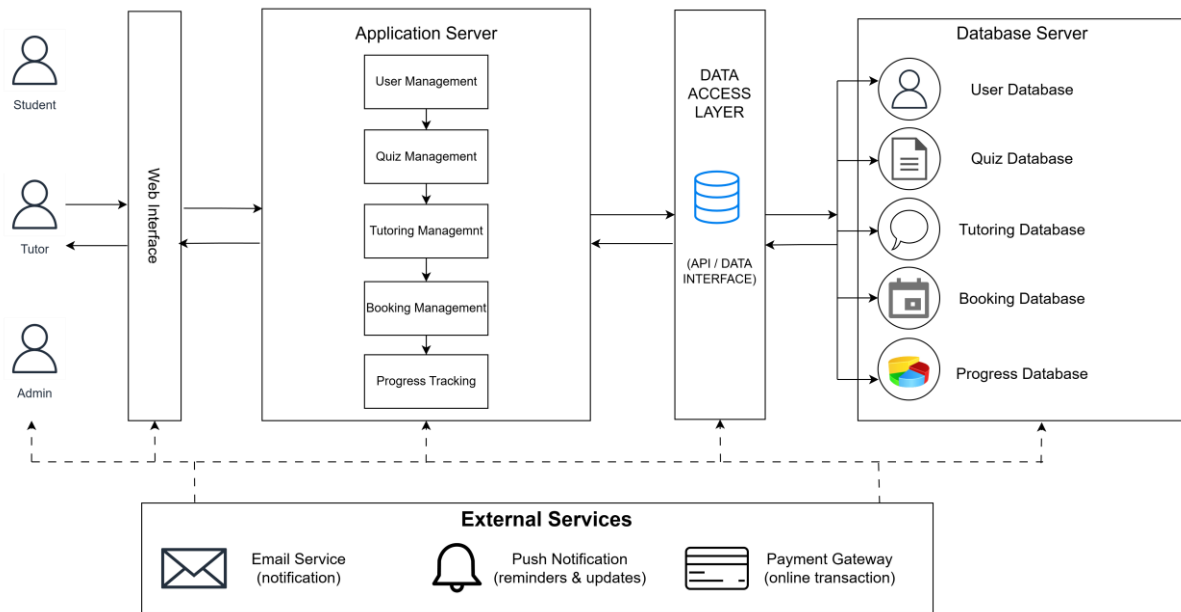
Prepared by:

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Section/Group: S1G1		
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1.0 SYSTEM ARCHITECTURE

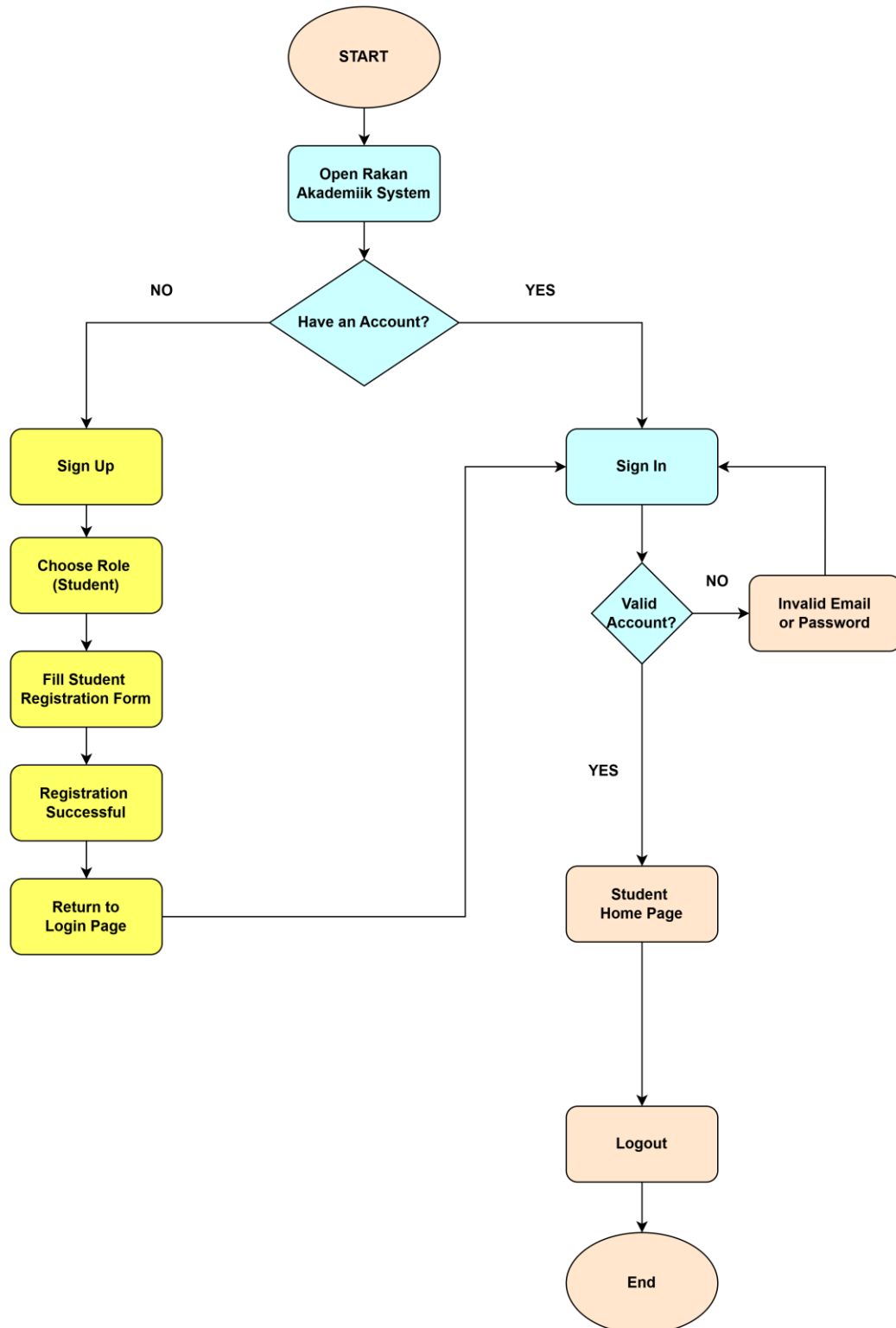


2.0 FUNCTION FLOWCHART

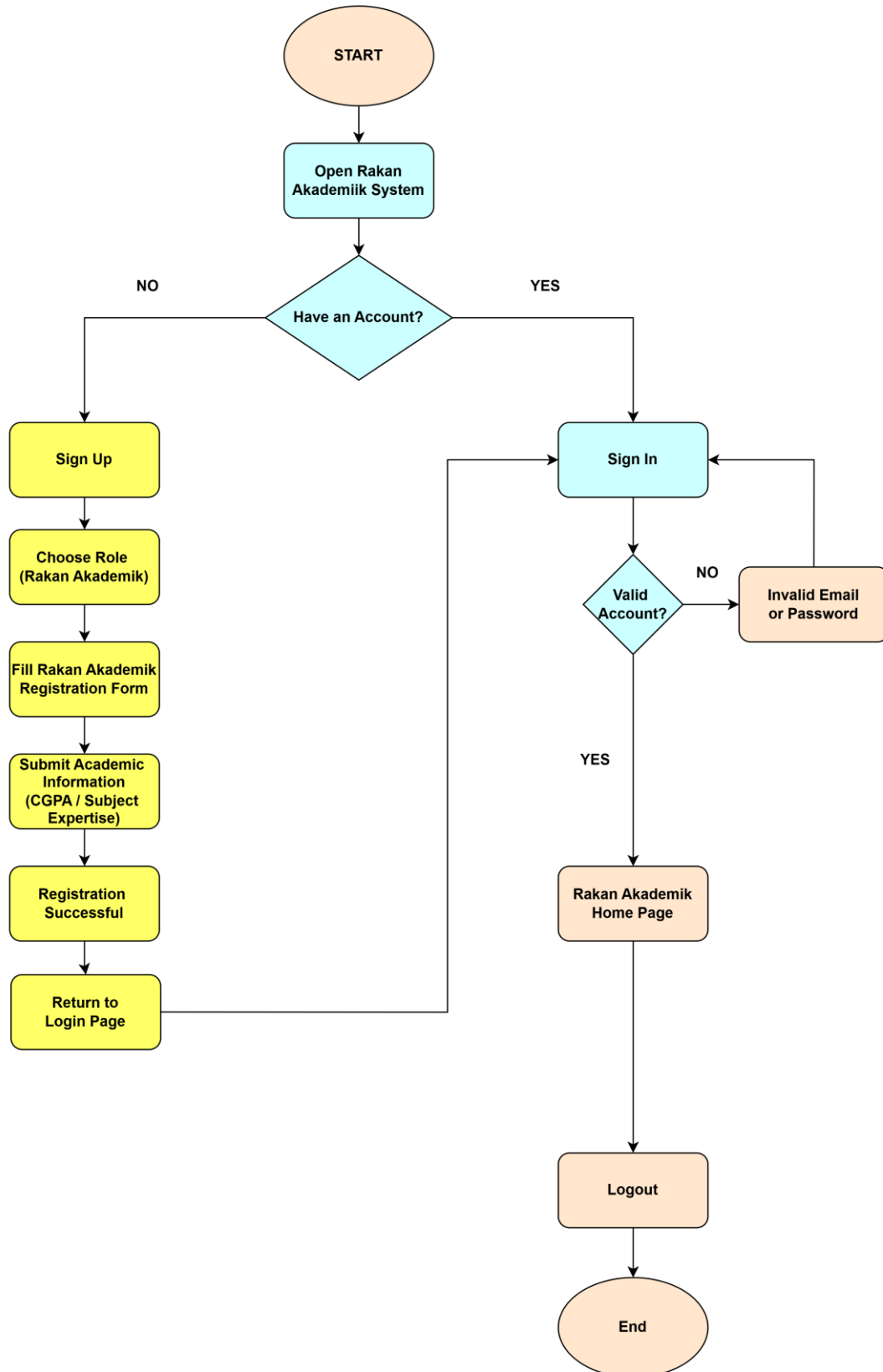
2.1 User Management System

– Student

1. STUDENT FLOWCHART

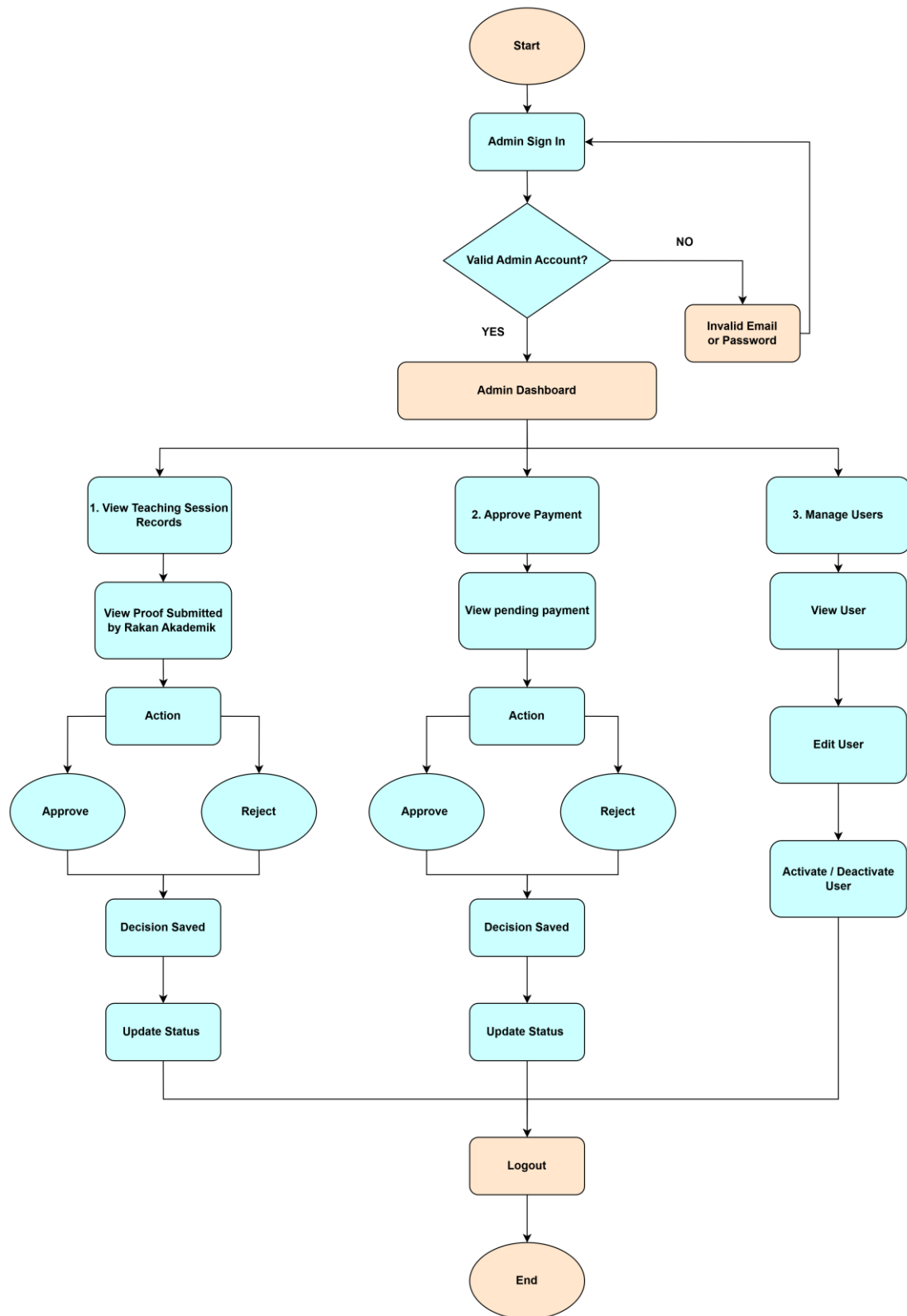


2. RAKAN AKADEMIK FLOWCHART



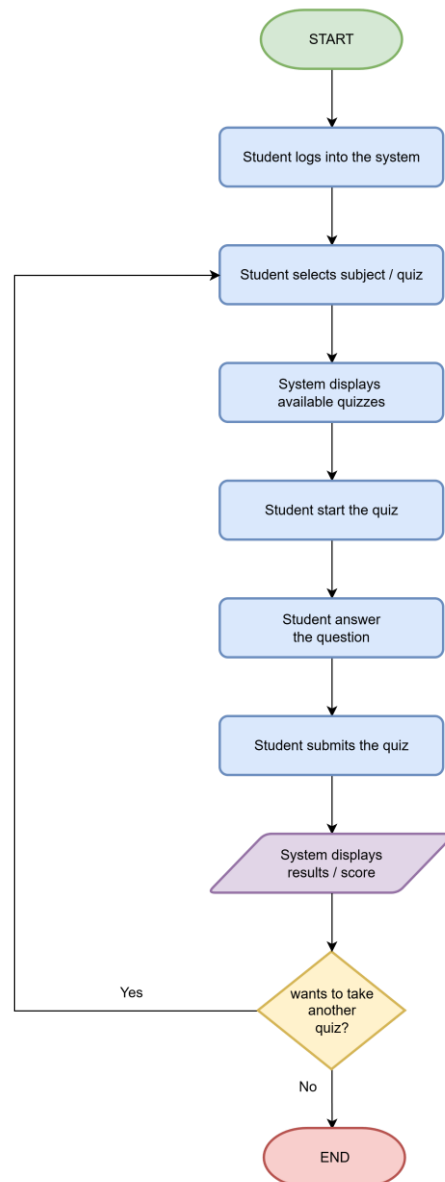
– Admin

3. ADMIN FLOWCHART

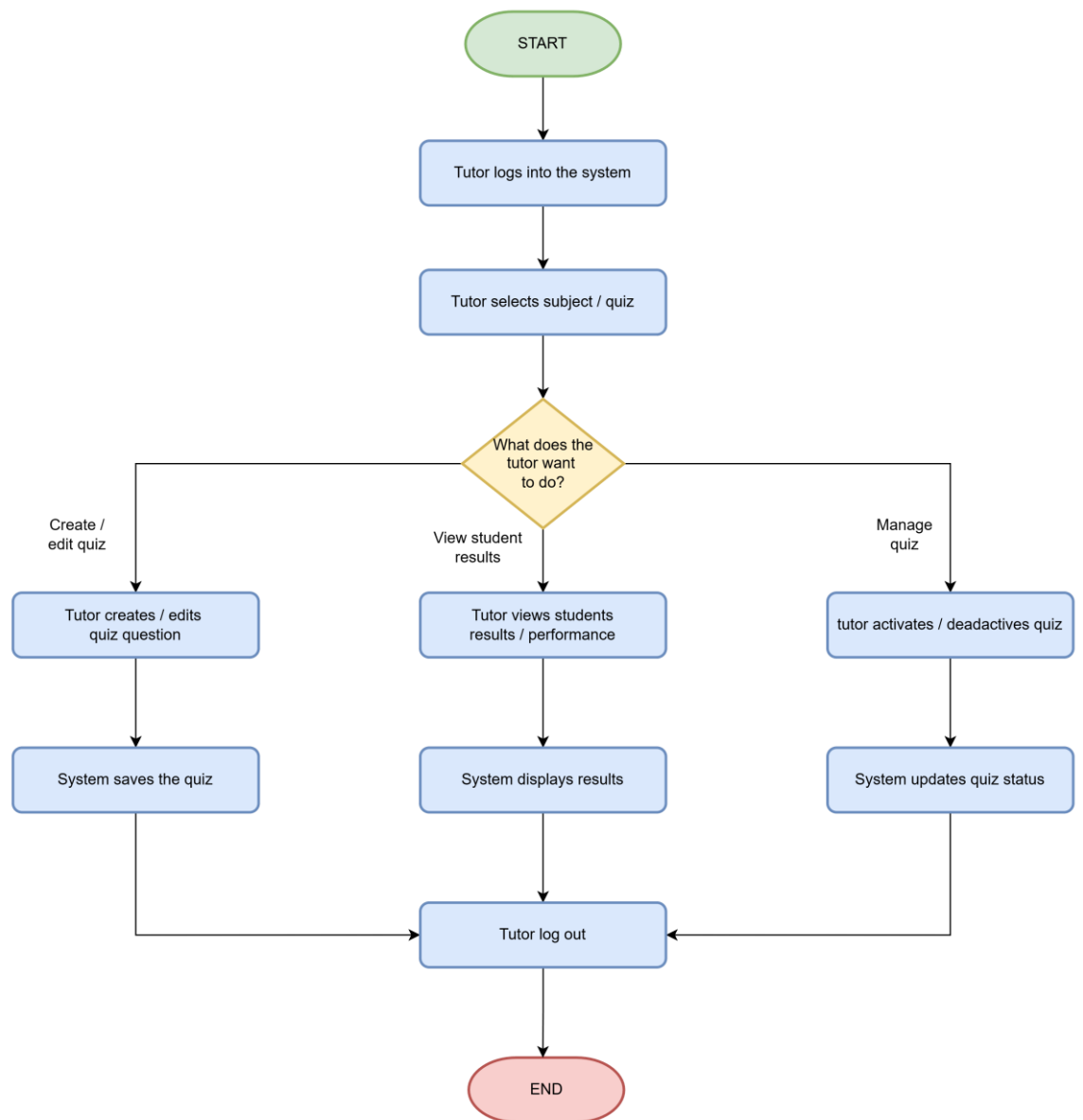


2.2 Quiz System

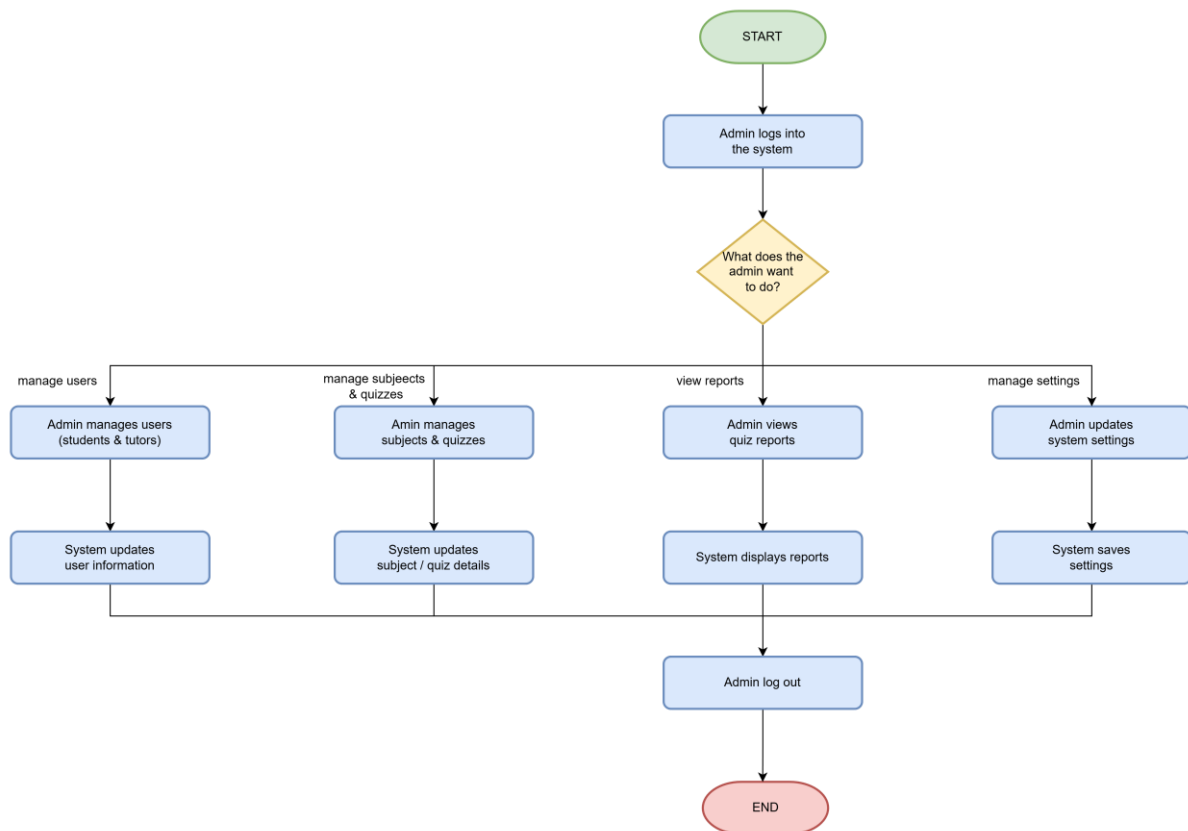
– Student



– Tutor

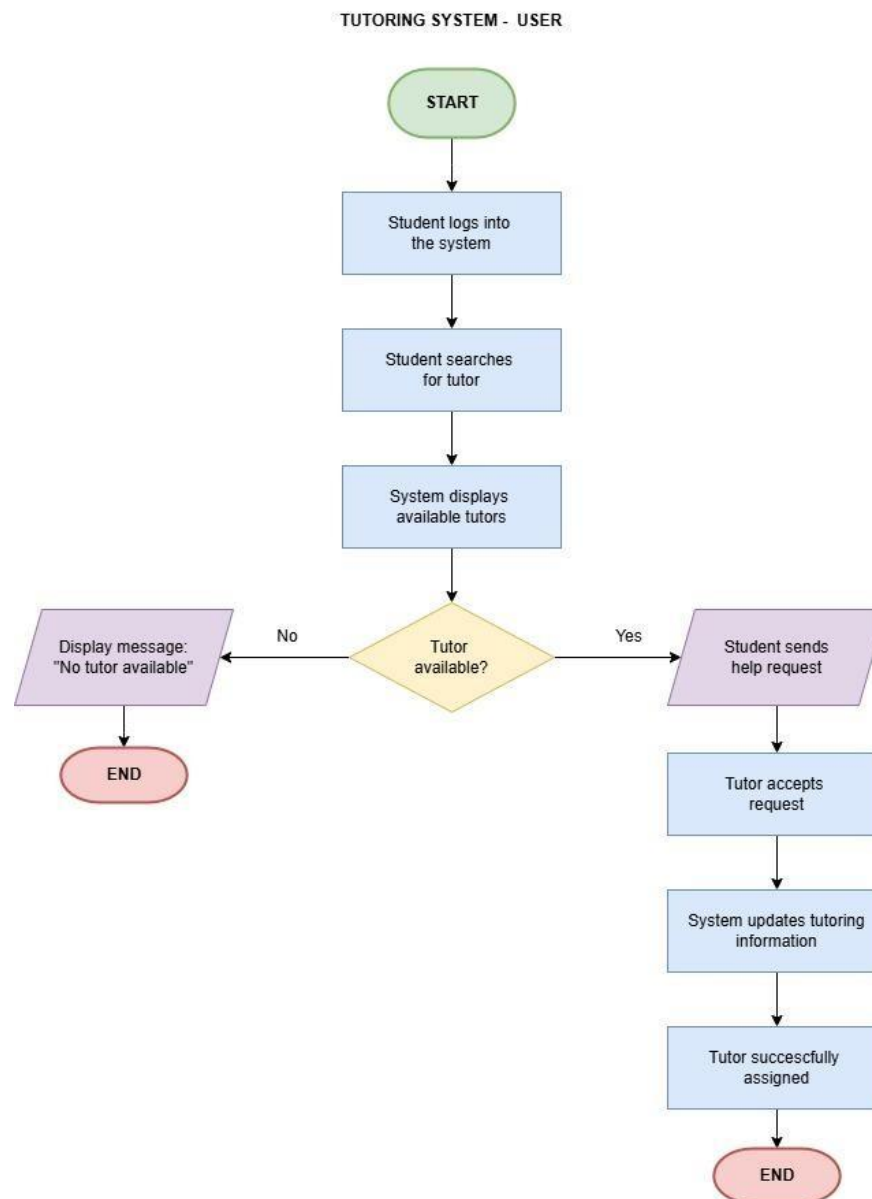


– Admin

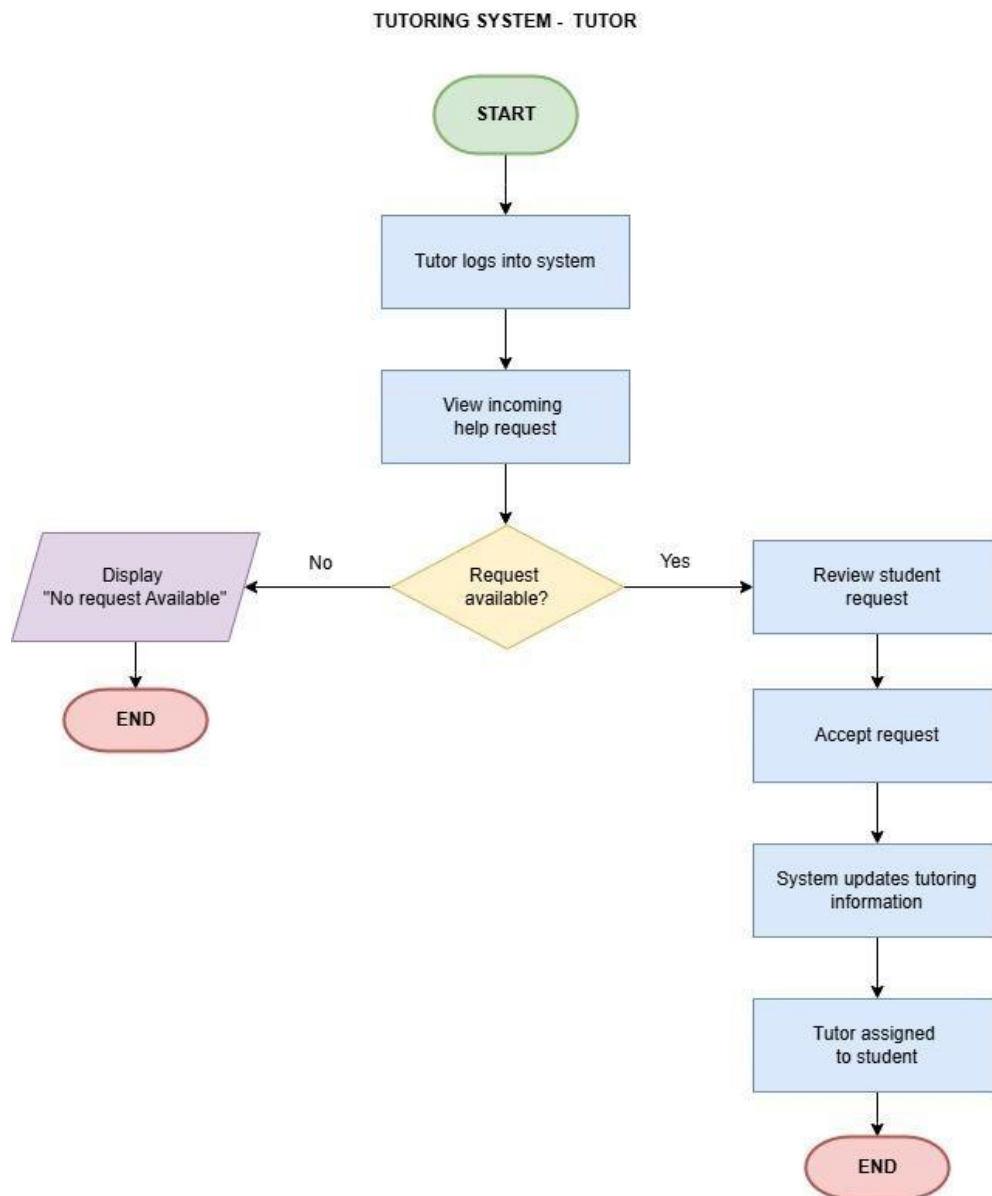


2.3 Tutoring System

– Student

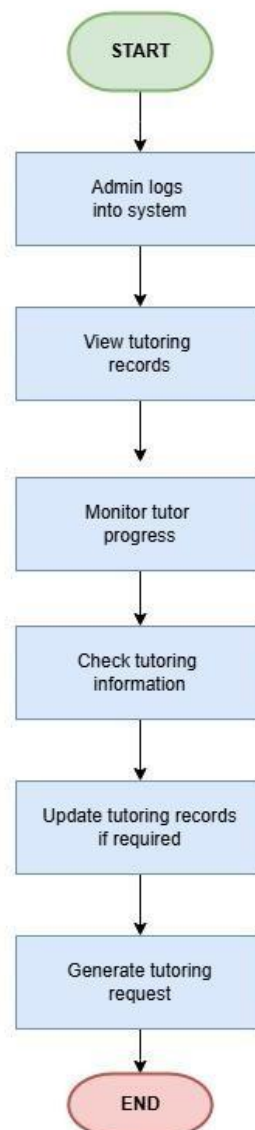


– Tutor



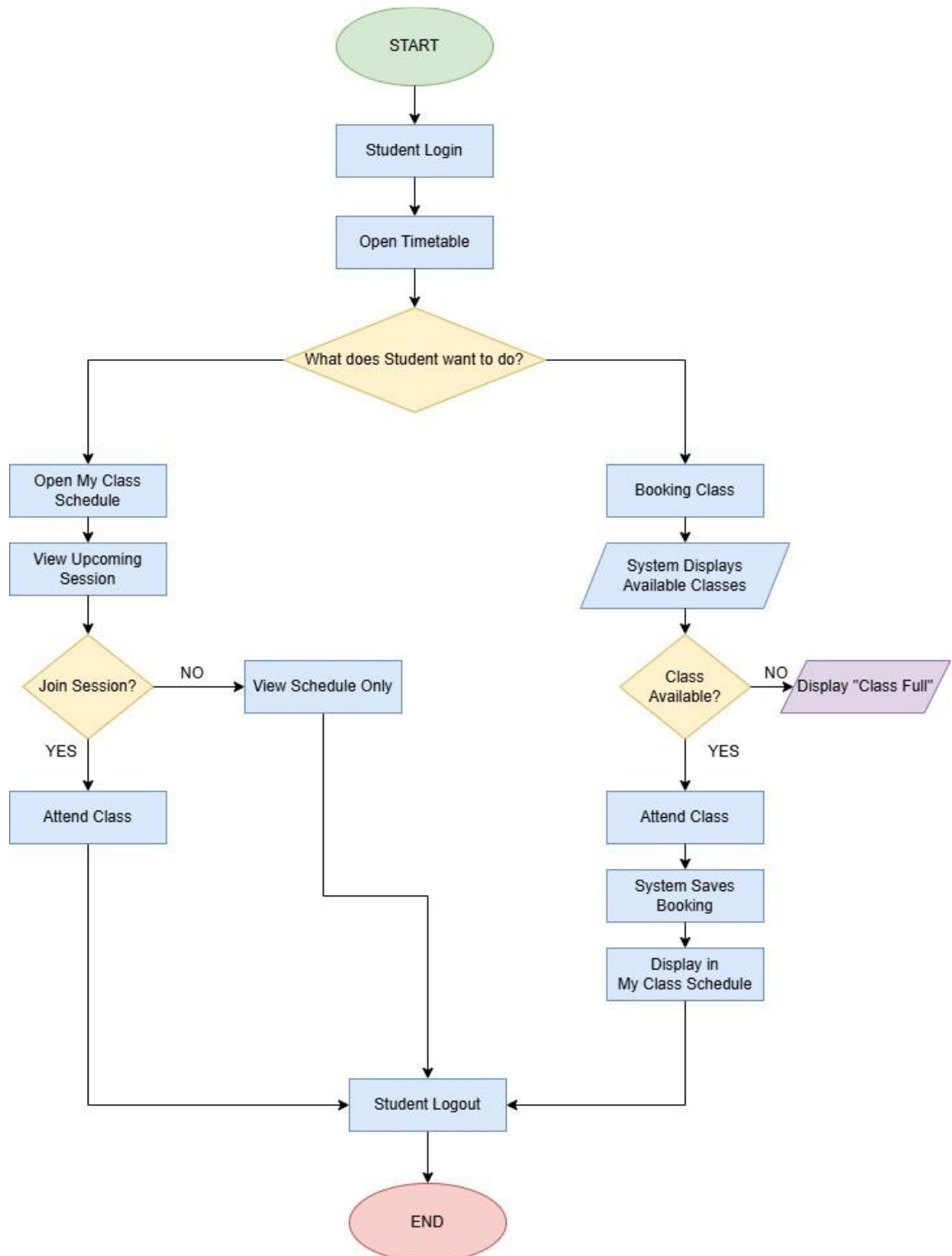
– Admin

TUTORING SYSTEM - ADMIN

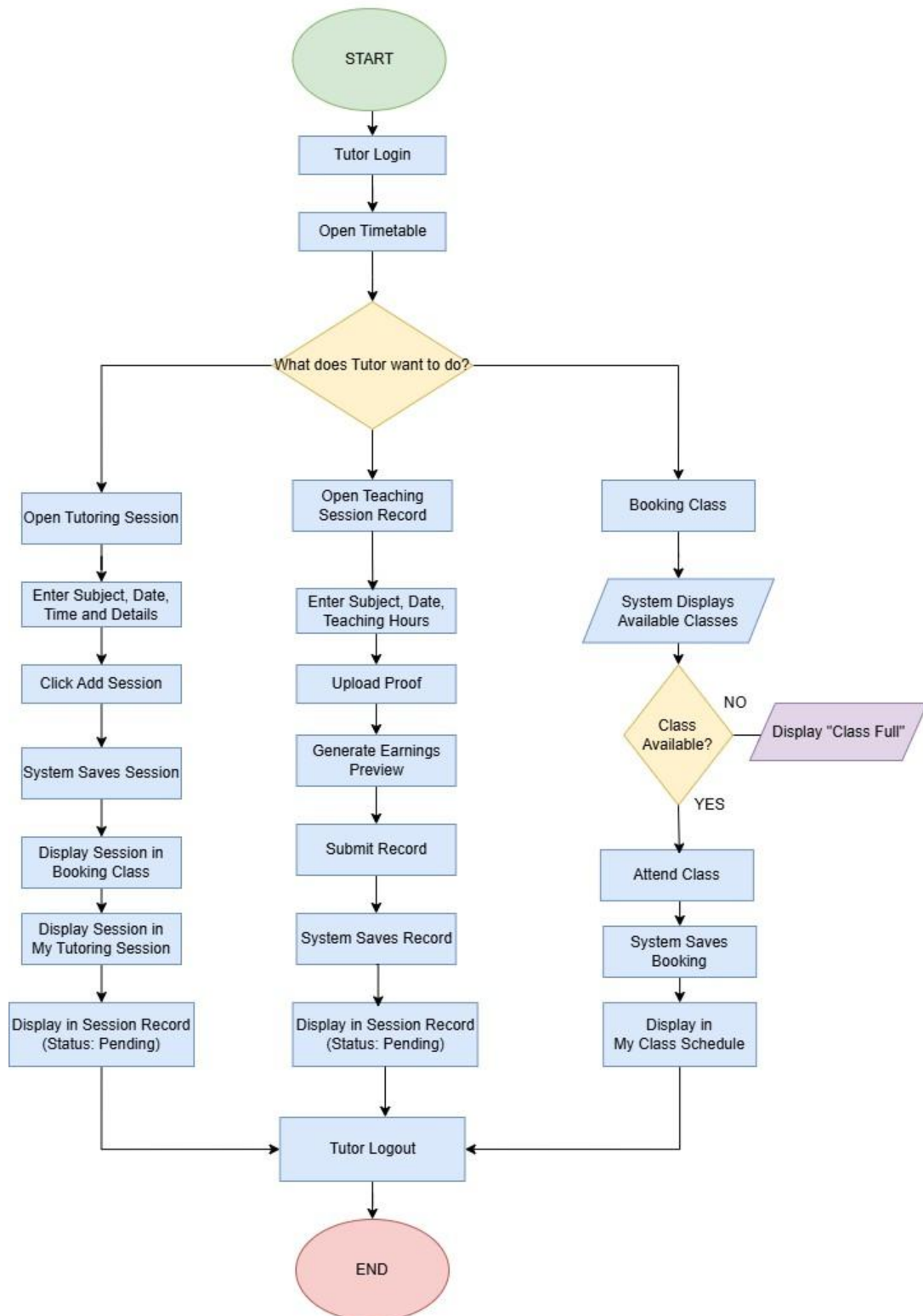


2.4 Booking System

– Student

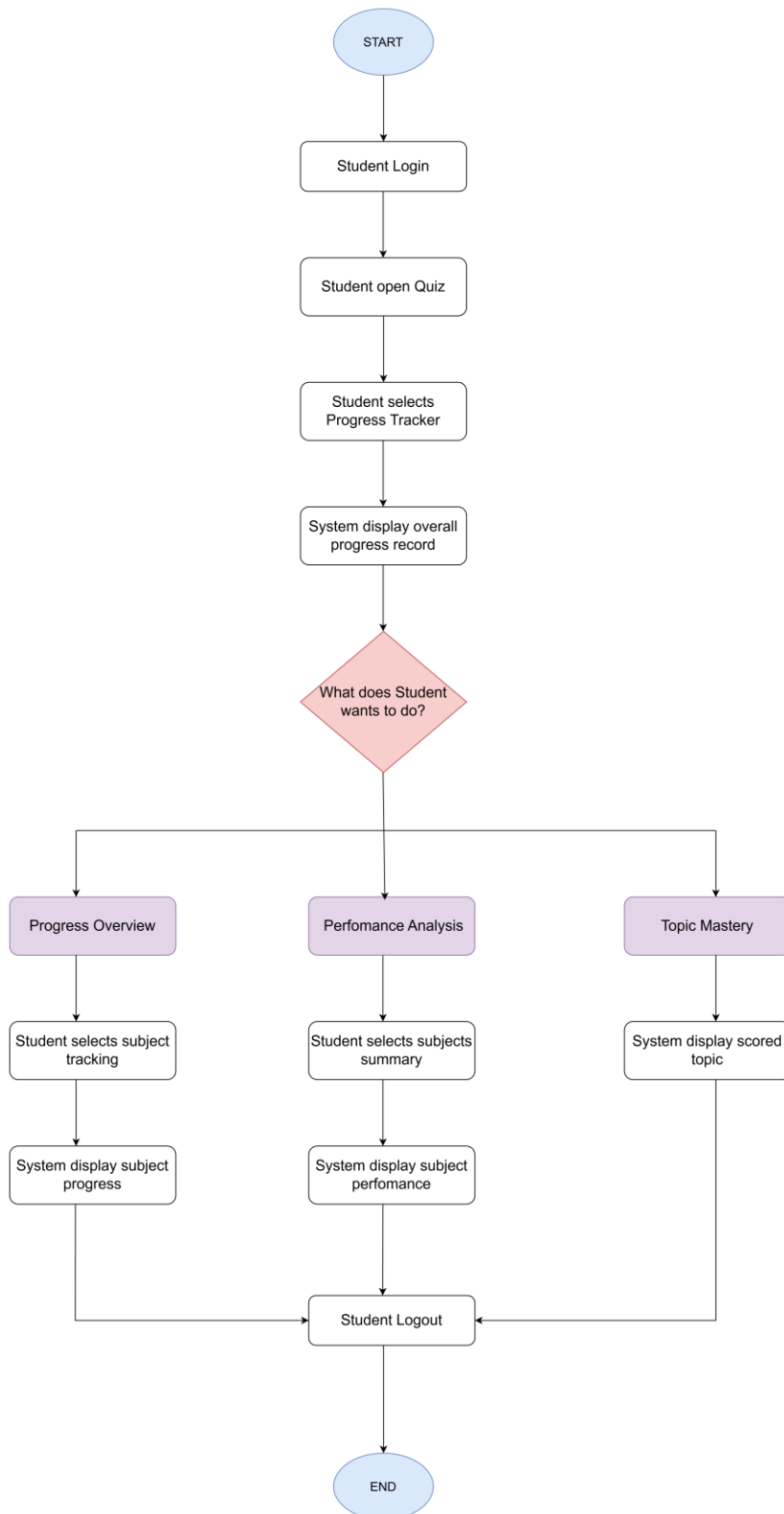


- Tutor

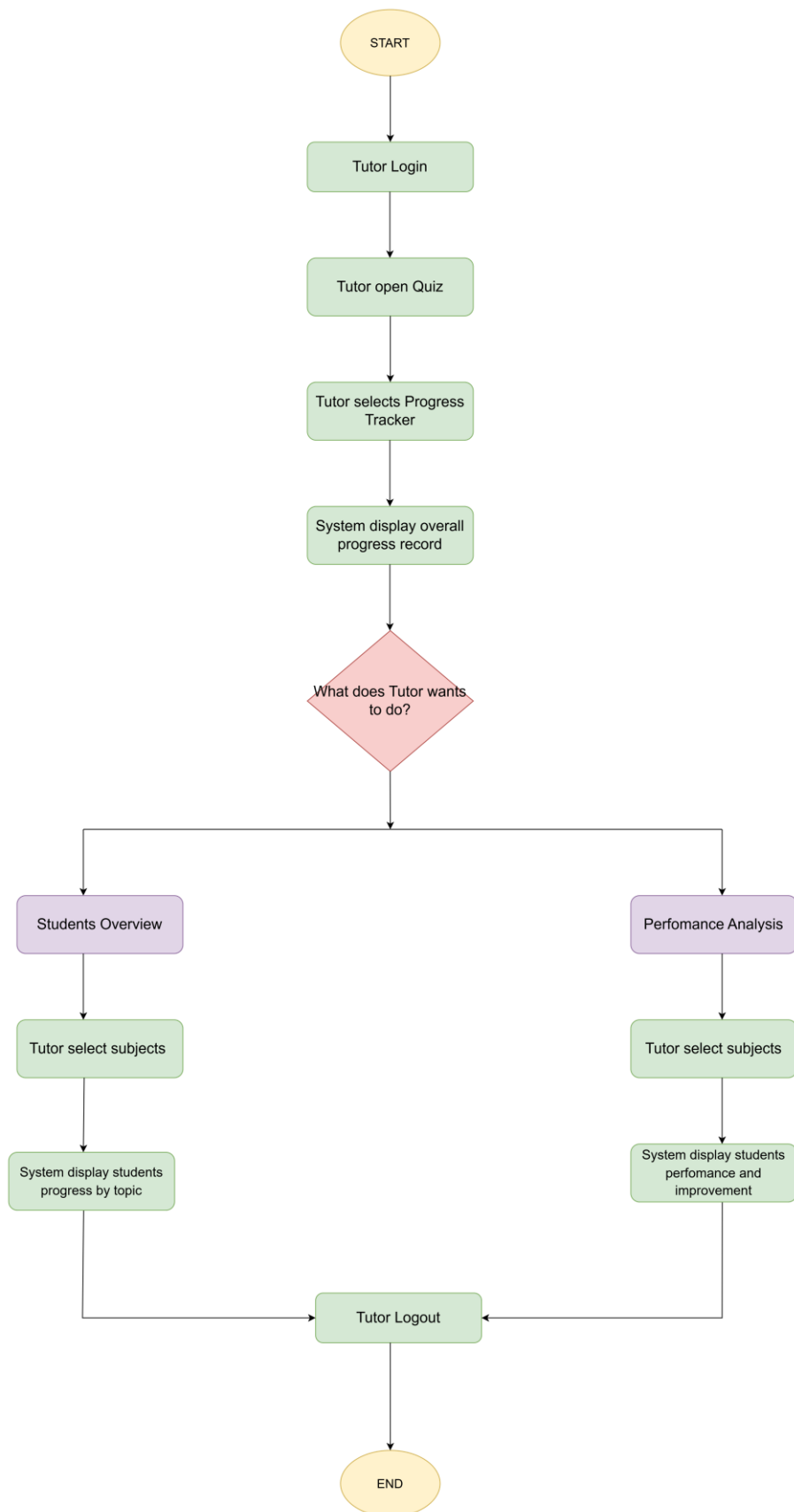


2.5 Progress Tracking System

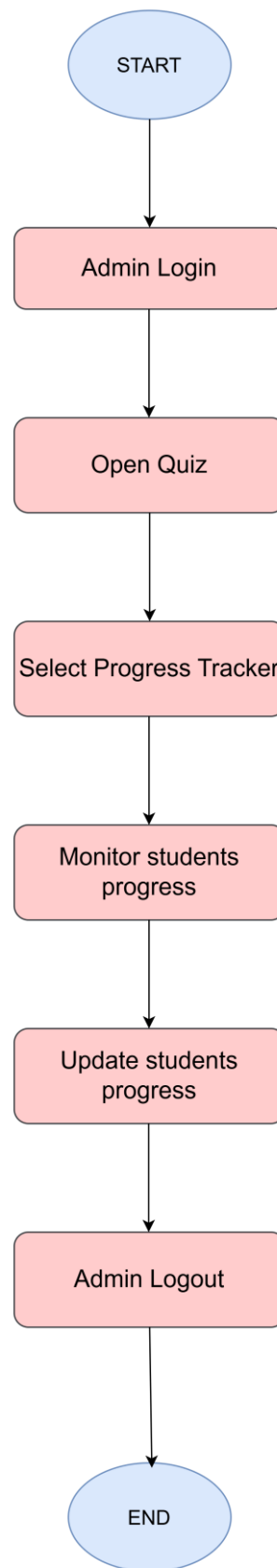
– Student



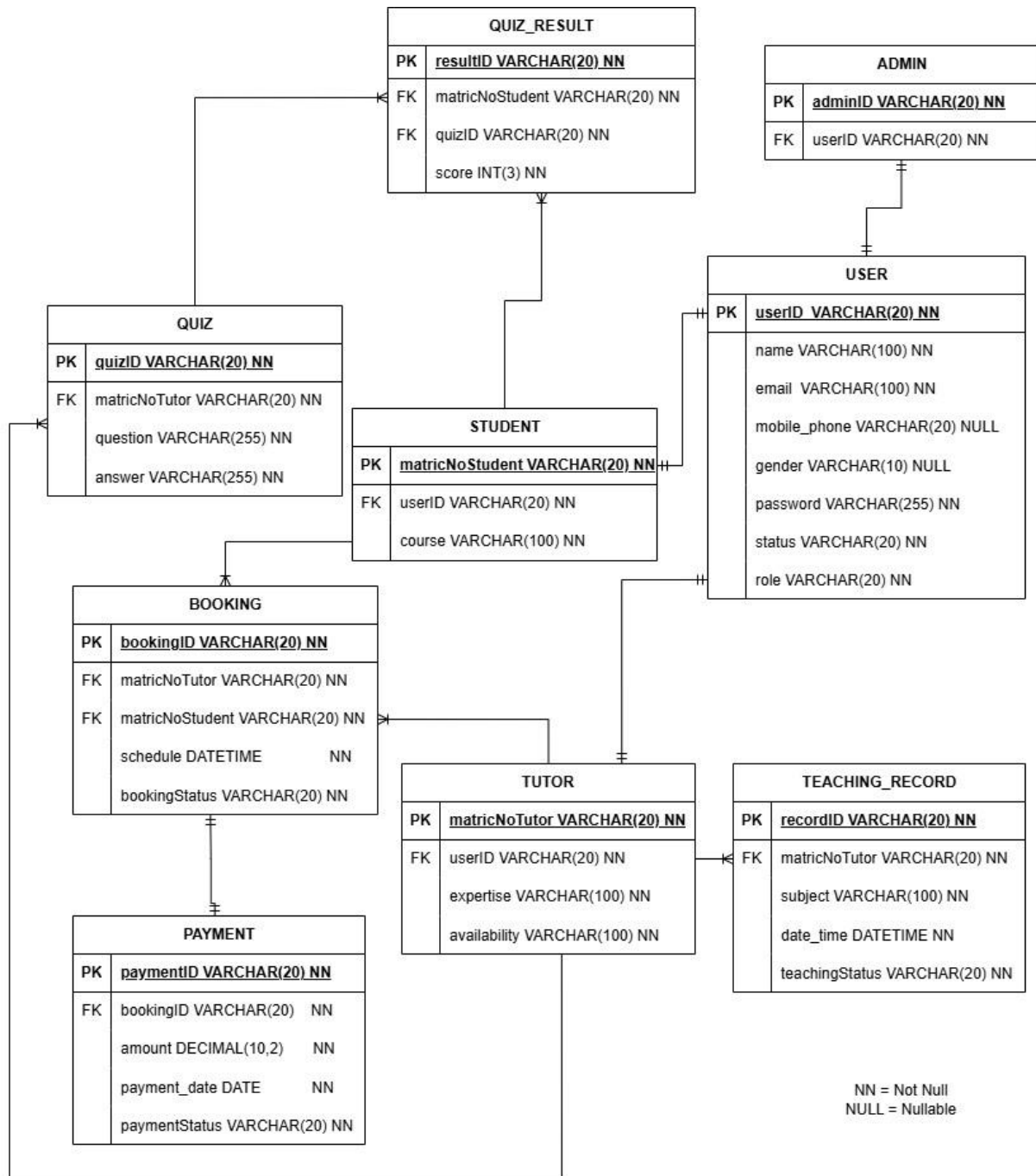
– Tutor



- Admin



3.0 PHYSICAL ERD



4.0 DATA DICTIONARY

1. User

Attribute	Data Type	Constraint	Description
userID	VARCHAR (20)	Primary Key Not Null	Unique user identifier
name	VARCHAR (100)	Not Null	User full name
email	VARCHAR (100)	Null	User email address
mobile_phone	VARCHAR (20)	Null	User phone number
gender	VARCHAR (10)	Not Null	User gender
password	VARCHAR (255)	Not Null	User password
status	VARCHAR (20)	Not Null	User account status
role	VARCHAR (20)	Not Null	User role

2. Student

Attribute	Data Type	Constraint	Description
matricNoStudent	VARCHAR (20)	Primary Key Not Null	Student matric number
userID	VARCHAR (20)	Foreign Key Not Null	Reference to User
course	VARCHAR (100)	Not Null	Student course

3. Tutor

Attribute	Data Type	Constraint	Description
matricNoTutor	VARCHAR (20)	Primary Key Not Null	Tutor matric number
userID	VARCHAR (20)	Foreign Key Not Null	Reference to User
expertise	VARCHAR (100)	Not Null	Tutor expertise
availability	VARCHAR (100)	Not Null	Tutor availability

4. Admin

Attribute	Data Type	Constraint	Description
adminID	VARCHAR (20)	Primary Key Not Null	Admin identifier
userID	VARCHAR (20)	Foreign Key Not Null	Reference to User

5. Booking

Attribute	Data Type	Constraint	Description
bookingID	VARCHAR (20)	Primary Key Not Null	Booking identifier
matricNoTutor	VARCHAR (20)	Foreign Key Not Null	Tutor assigned
matricNoStudent	VARCHAR (20)	Foreign Key Not Null	Student making booking
schedule	DATETIME	Not Null	Booking schedule
bookingStatus	VARCHAR (20)	Not Null	Booking status

6. Payment

Attribute	Data Type	Constraint	Description
paymentID	VARCHAR (20)	Primary Key Not Null	Payment identifier
bookingID	VARCHAR (20)	Foreign Key Not Null	Reference to booking
amount	DECIMAL (10,2)	Not Null	Payment amount
payment_date	DATE	Not Null	Payment date
paymentStatus	VARCHAR (20)	Not Null	Payment status

7. Quiz

Attribute	Data Type	Constraint	Description
quizID	VARCHAR (20)	Primary Key Not Null	Quiz identifier
matricNoTutor	VARCHAR (20)	Foreign Key Not Null	Tutor who created quiz
question	VARCHAR (255)	Not Null	Quiz question
answer	VARCHAR (255)	Not Null	Correct answer

8. Quiz Result

Attribute	Data Type	Constraint	Description
resultID	VARCHAR (20)	Primary Key Not Null	Quiz result identifier
matricNoStudent	VARCHAR (20)	Foreign Key Not Null	Student taking quiz
quizID	VARCHAR (20)	Foreign Key Not Null	Reference to quiz
score	INT (3)	Not Null	Quiz score

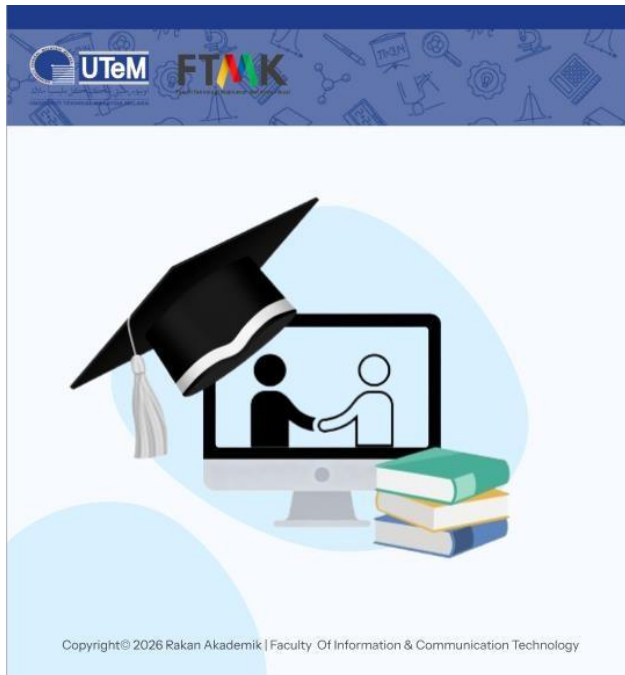
9. Teaching Record

Attribute	Data Type	Constraint	Description
recordID	VARCHAR (20)	Primary Key Not Null	Teaching record identifier
matricNoTutor	VARCHAR (20)	Foreign Key Not Null	Reference to tutor
subject	VARCHAR (100)	Not Null	Subject taught
date_time	DATETIME	Not Null	Session date and time
teachingStatus	VARCHAR (20)	Not Null	Teaching session status

5.0 USER INTERFACE DESIGN

– Student

Log in / Sign up



Student Id

Password 

☐ Remember me [Forgot Password?](#)

[Don't have an account?](#)



Sign Up As Student

Name

Email

Mobile Phone

Gender
☐ Male ☐ Female

Password

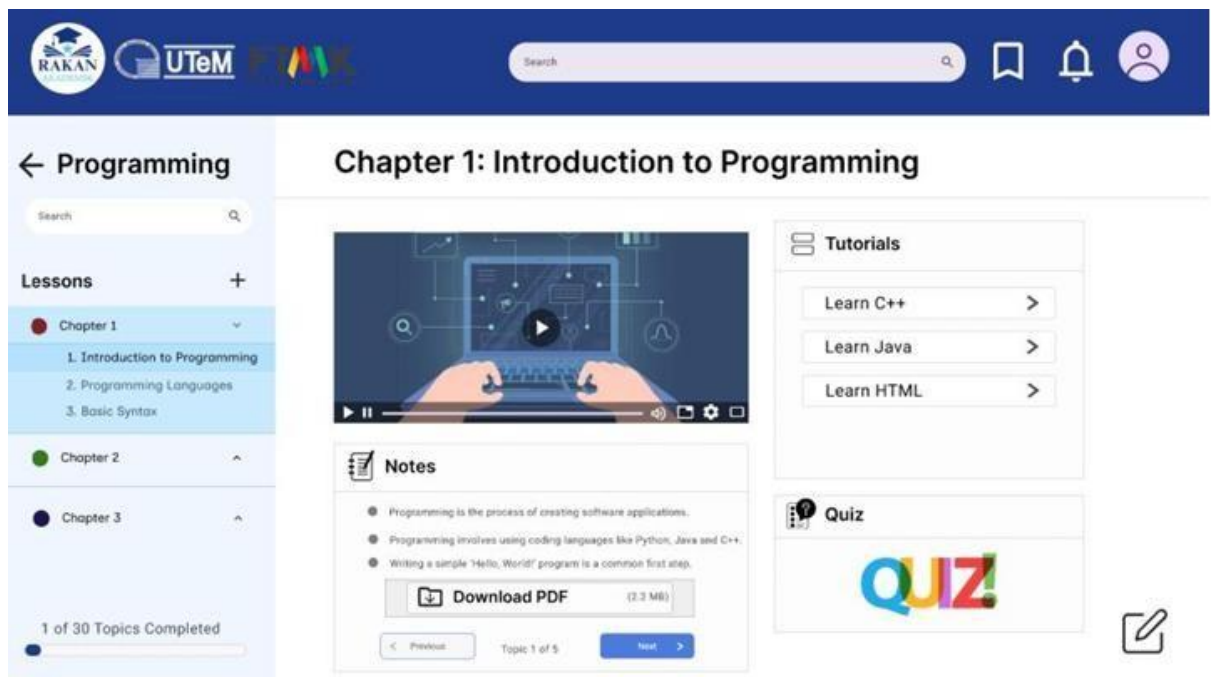
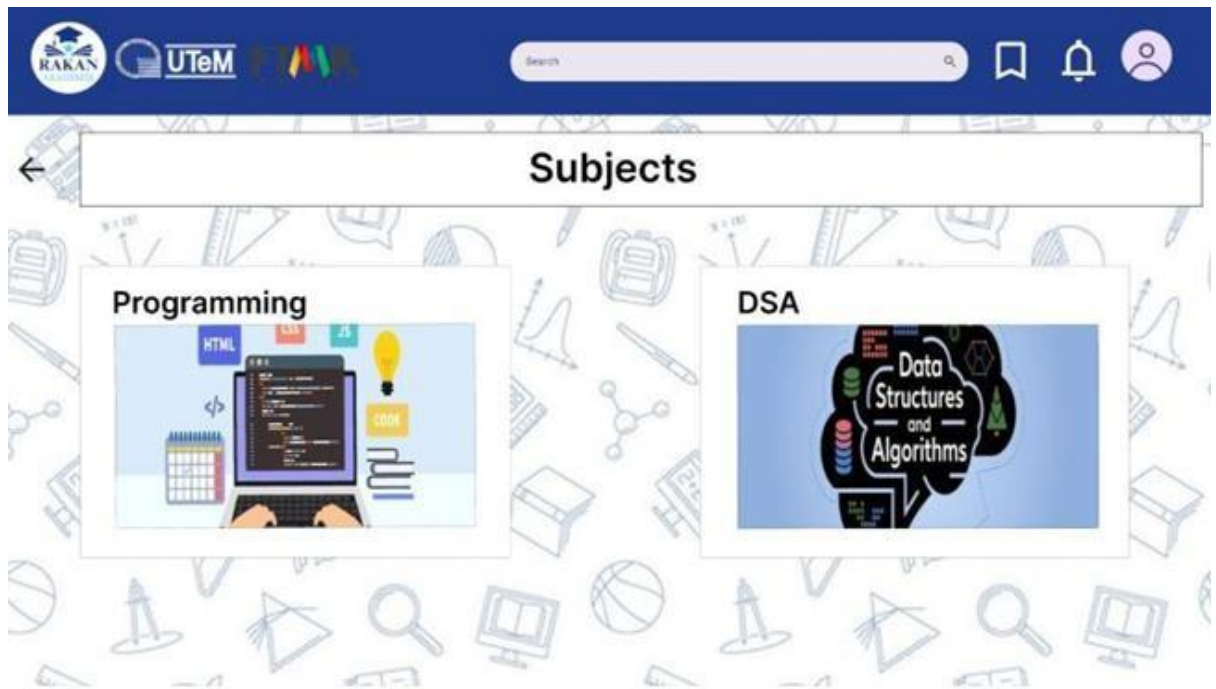
Confirm Password

Main page

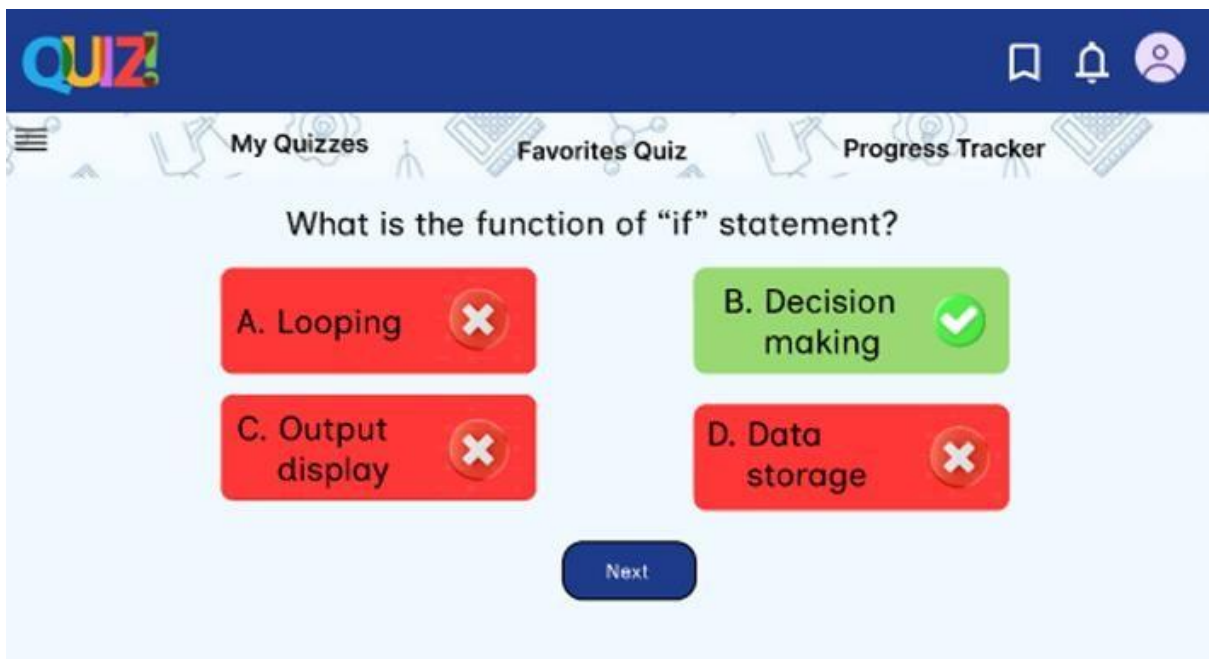


Learning module





Quiz



- Home
- Progress Overview
- Performance Analysis
- Topic Mastery

Quizzes

Favorites Quiz

Progress Tracker

Your Progress

All classes>

72%

Average mark

67%

Activity

80%

Attendance

Subject

Data Structure Algorithm

70%

View Details →

Programming

50%

View Details →

Recent Activity

Quiz Completed

Arrays - DSA

78%

Quiz Completed

Linked List- DSA

65%

Question Bank

Session Joined

Programming Tutoring

- Home
- Progress Overview
- Performance Analysis
- Topic Mastery

Performance Analysis

Analyze your quiz performance over time and identify areas of improvement.

All Subjects ▾

Average Mark

72%

Quizzes Completed

24/40

This Semester

Improvement

+12%

Great Progress

Score Trend

Last 8 Weeks ▾

Week	Score
Week 1	20
Week 2	40
Week 3	50
Week 4	60
Week 5	70
Week 6	72%
Week 7	80
Week 8	100

Subject Summary

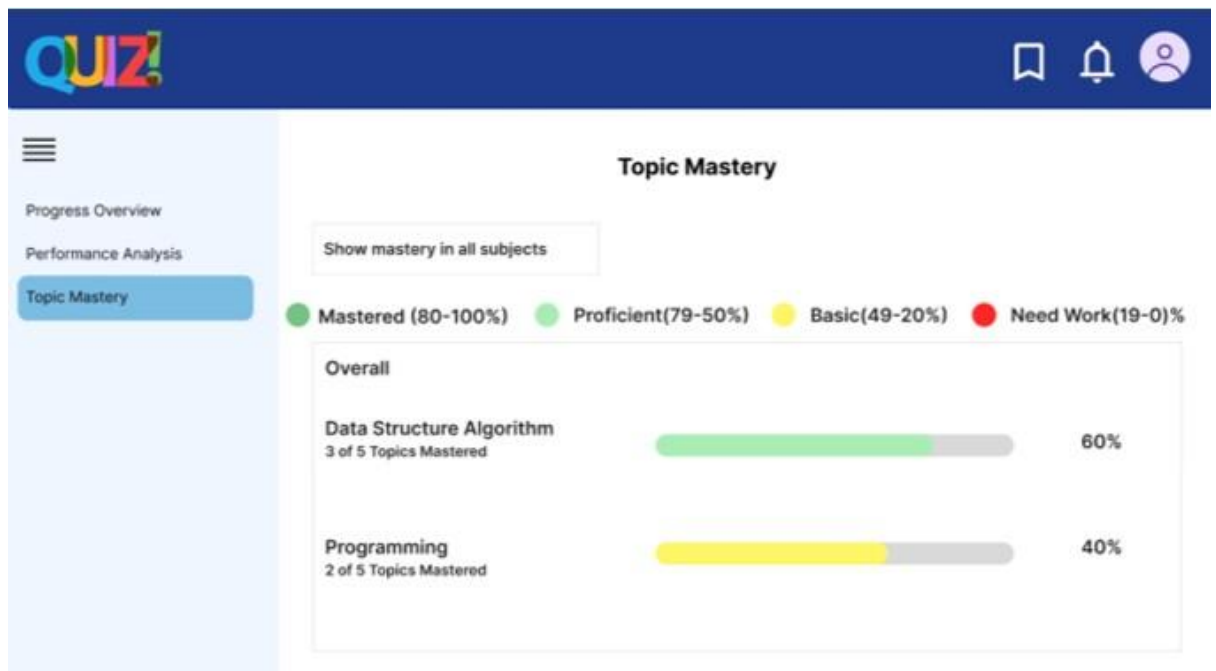
Data Structure Algorithm

+10% improvement

Programming

+14% improvement

View All →



QUIZ!

Home
Tutor Matching
Request Help
Discussion
Feedback

Tutor Matching

5
Rakan Akademik

3
Matched Sessions

1
Pending Approval

Find Tutor

Select Subject: Programming
Select Date: Tomorrow
Select Sessions: 10.00-11.00 AM
Search

Available Tutor

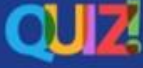
Ahmad Rayyan Programming Tutor	10.00-11.00 AM	Match
Puteri Damia Programming Tutor	10.00-11.00 AM	Matched

Matching Requested

Puteri Damia
Programming Tutor

Tomorrow,
10.00-11.00 AM

Matched





Home

Tutor Matching

Request Help

Discussion

Feedback

Request Help

Select Subject

Select Topic

Message

Request Help





Home

Tutor Matching

Request Help

Discussion

Feedback

Discussion/Chat

Select a subject and tutor to start a discussion.

Select Subject

Topic (optional)

Search Tutor

Available Tutors

	Puteri Damia Programming Tutor Arrays	Start Discussion
	Haikal Data Structure Tutor Queue	Start Discussion
	Ahmad Programming Tutor Arrays	Start Discussion
	Sarah Programming Tutor Arrays	Start Discussion

[My Quizzes](#)
[Question Bank](#)
[Progress Tracker](#)

Home

Tutor Matching

Request Help

Discussion

Feedback

Feedback

Feedback from tutoring session.

Select Tutor (optional)

All subjects

Search Feedback

Puteri Damia
Programming Tutor

Today, 12:00PM

The feedback...

Ahmad
Data Structure Tutor

Today, 10:00AM

The feedback...

Haikal
Programming Tutor

Yesterday, 1:00PM

The feedback...

Timetable

Student

Booking Class

My Class Schedule

Logout

Book Tutoring Session

[Upcoming](#)
[Past](#)

TIME	EVENT TIME	TUTOR NAME	ACTION	
16-APRIL-2026 03:00 PM - 04:00 PM	DATA STRUCTURE AND ALGORITHM	NURSYHADAH	ATTEND	7/10
20-APRIL-2026 02:00 PM - 03:00 PM	PROGRAMMING	AHMAD ALI	ATTEND	6/10
22-APRIL-2026 04:00 PM - 05:00 PM	DATA STRUCTURE AND ALGORITHM	AHMAD ABU	ATTEND	2/10
27-APRIL-2026 02:00 PM - 03:00 PM	PROGRAMMING	NURSYHADAH	ATTEND	10/10
29-APRIL-2026 03:00 PM - 04:00 PM	PROGRAMMING	AHMAD ALI	ATTEND	8/10

Student

Booking Class

My Class Schedule

Logout

My Class Schedule

Upcoming Session		Completed Session	
DATE/TIME	EVENT TIME	TUTOR NAME	ACTION
16-APRIL-2026 03:00 PM - 04:00 PM	DATA STRUCTURE AND ALGORITHM	NURSYUHADAH	JOIN
20-APRIL-2026 02:00 PM - 03:00 PM	PROGRAMMING	AHMAD ALI	JOIN
29-APRIL-2026 03:00 PM - 04:00 PM	PROGRAMMING	AHMAD ALI	JOIN

Find Tutor

Home

Rakan Akademik

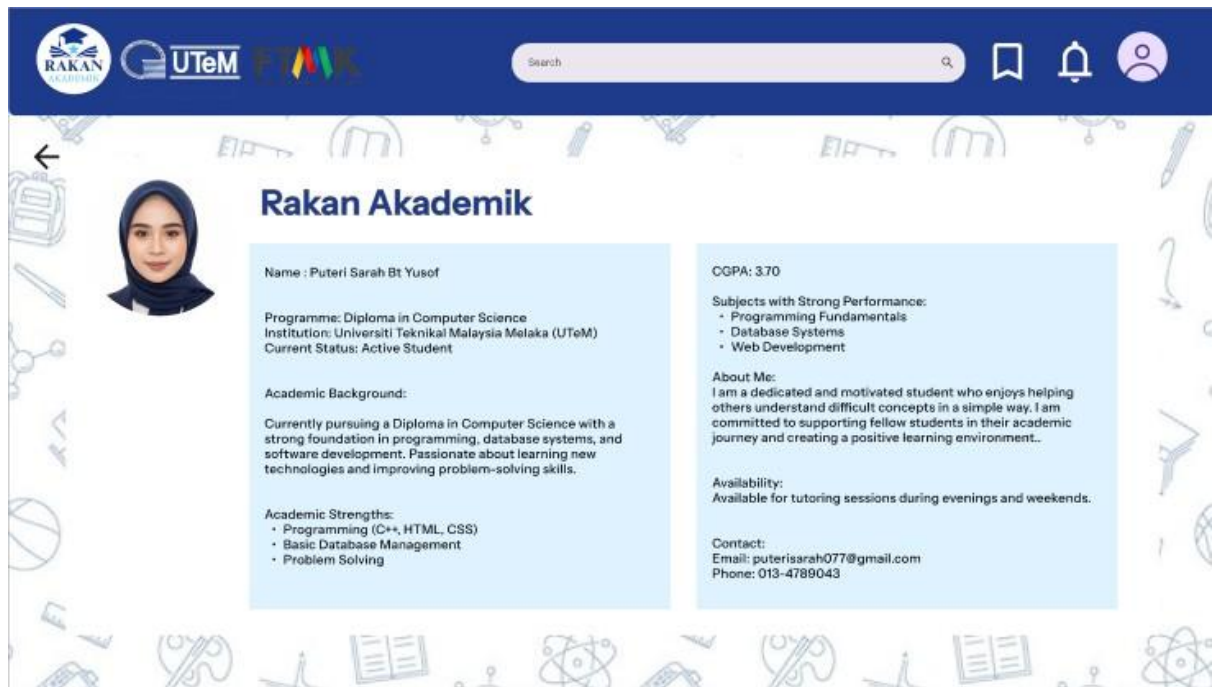
Kindly, please choose Rakan Akademik :

Noor Aisya Sofea Bt Ahmad Jamal

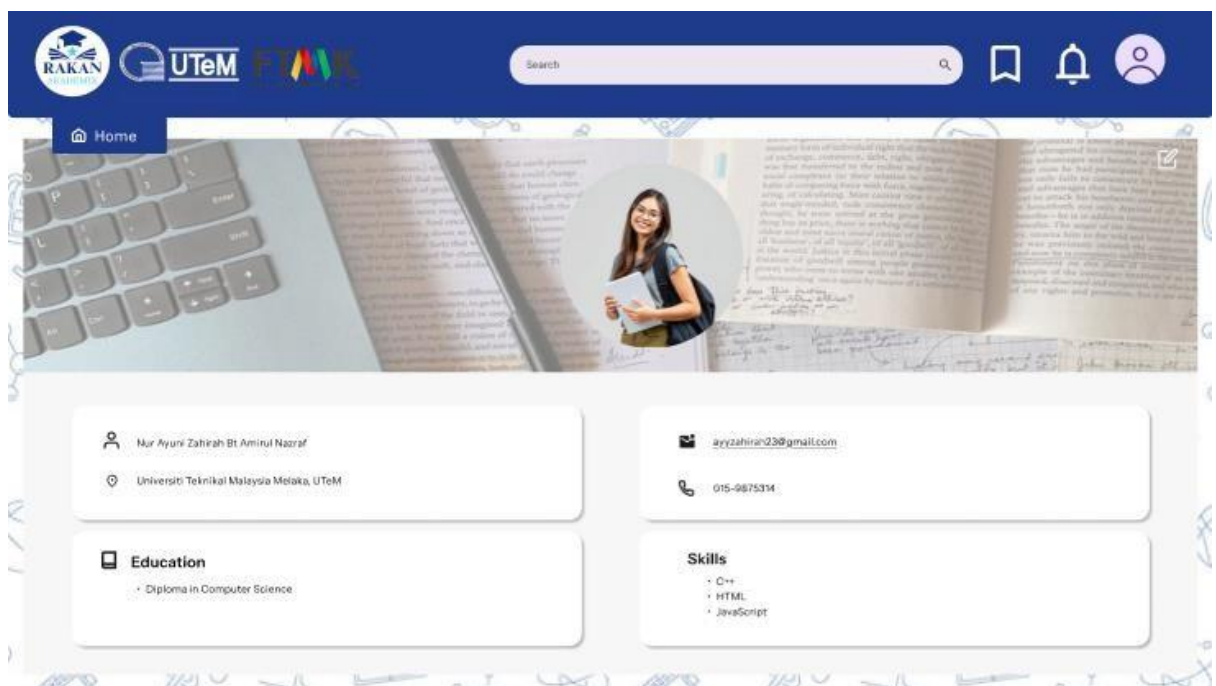
Ali Bin Umair

Harraz Bin Husin

Puteri Sarah Bt Yusof

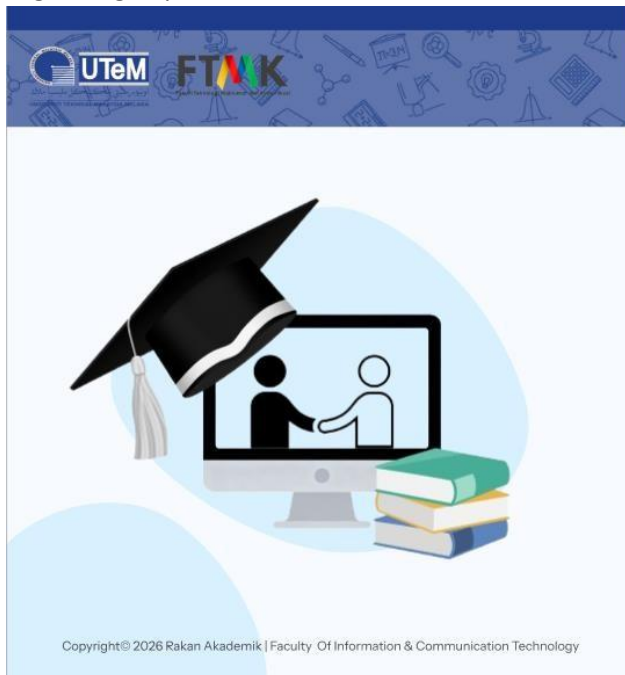


Profile page



– Rakan Akademik

Login / Sign up



Student Id

Password

☐ Remember me [Forgot Password?](#)

[Log in](#)

[Don't have an account?](#) [Sign up](#)



Sign Up As Student

Name

Email

Mobile Phone

Gender

☐ Male ☐ Female

Password

Confirm Password

[Sign up](#)

Profile page



Search

[Home](#)



**Puteri Sarah Bt Yusof**
 Universiti Teknikal Malaysia Melaka, UTeM

 puterisarah077@gmail.com
 013-4789043

**Education**

- CGPA: 3.70
- Bachelor in Artificial Intelligence
- Diploma in Computer Science

Skills

- CSS
- HTML
- JavaScript



Search





Rakan Akademik

Name : Puteri Sarah Bt Yusof

Programme: Diploma in Computer Science
Institution: Universiti Teknikal Malaysia Melaka (UTeM)
Current Status: Active Student

Academic Background:

Currently pursuing a Diploma in Computer Science with a strong foundation in programming, database systems, and software development. Passionate about learning new technologies and improving problem-solving skills.

Academic Strengths:

- Programming (C++, HTML, CSS)
- Basic Database Management
- Problem Solving

CGPA: 3.70

Subjects with Strong Performance:

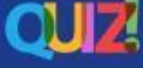
- Programming Fundamentals
- Database Systems
- Web Development

About Me:
I am a dedicated and motivated student who enjoys helping others understand difficult concepts in a simple way. I am committed to supporting fellow students in their academic journey and creating a positive learning environment.

Availability:
Available for tutoring sessions during evenings and weekends.

Contact:
Email: puterisarah077@gmail.com
Phone: 013-4789043

Quiz



**My Quizzes****Favorites Quiz****Progress Tracker**

choose one from categories below

Programming

Data Structure & Algorithm



Review & Publish
design engaging quizzes for students

1 Quiz Details

2 Questions

3 Quiz Settings

Quiz Overview

**Introduction to C++ Programming**
Test your knowledge on C++ basic, syntax and fundamentals
⌚ 30 minutes ↺ 2 attempts allow medium level

 Edit Details

Settings & Rules

 Edit Settings

 **Time Limit**
30 minutes

 **Show Results**
Show immediately after submission

 **Attempts Allowed**
2 attempts

 **Visibility**
Private - Only invited students



Previous

Review & Publish

Create New Quiz

design engaging quizzes for students

1 Quiz Details 2 Questions 3 Quiz Settings

Quiz Title *

Introduction to C++ Programming

Description

Test your knowledge on C++ basic, syntax and fundamentals

Subject/Category

Programming

Difficulty Level

Easy

Medium

Hard

Quiz Cover (optional)



Introduction to C++
Test your knowledge on C++
Basic, syntax and fundamentals

save

☰ Drafts



PROGRAMMING



DATA STRUCTURE
AND ALGORITHM



Review & Publish

Drafts

DATA STRUCTURE AND ALGORITHM

What is graph?

answer A

Which data structure uses FIFO (First In First Out)?

answer C

Previous

Review & Publish

Home

Students Overview

Performance Analysis

Student Progress Overview

78%

Average mark

72%

Activity

85%

Attendance

8

Pending Tasks

Quizzes completed

128/160

+34 Sessions Conducted

upcoming sessions : 3

Student Progress by Topics

Filter by Topics: Data Structure Algorithm Array

Rank	Student	Topics	Average mark	Quizzes Completed
1	Maryam	Array	82% (10/10)	88%
2	Sarah	Array	59% (8/10)	72%
3	Kamilia	Array	55% (7/10)	60%

Recent Activity

Quiz Completed

Maryam | Arrays - DSA

75%

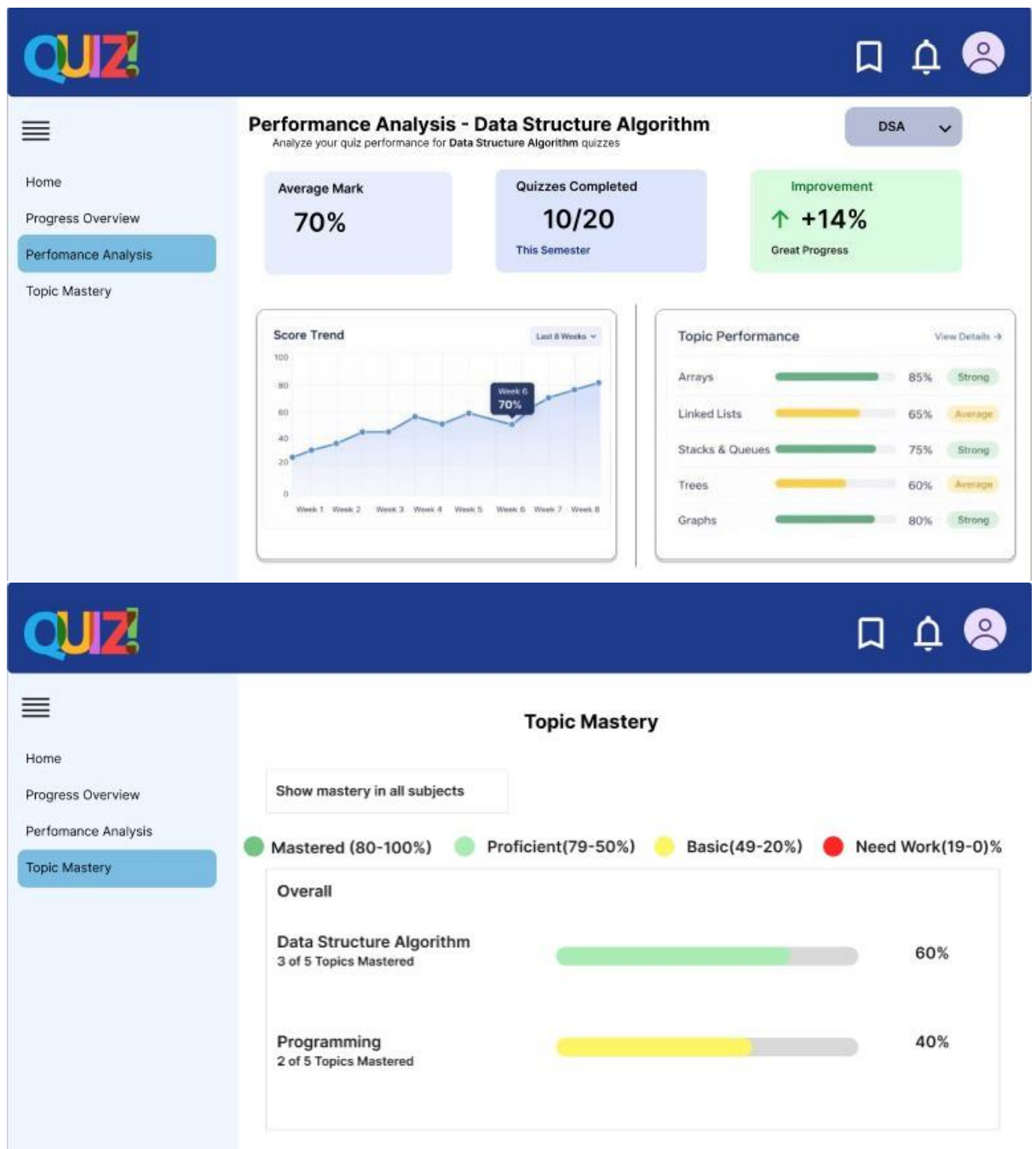
Quiz Completed

Aqil | Linked List- DSA

62%

Session Scheduled

DSA Tutoring- Today 1:00PM



Add tutoring











- Student/Tutor
- Booking Class
- My Class Schedule
- My Tutoring Sessions
- Add Tutoring Session**
- Add Teaching Session
- Session Records
- Earnings Dashboard
- Logout

Add Tutoring Session

Subject

^

Programming

Data Structure and Algorithm

UU

MM

YYYY

Start time

v

End time

v

ADD SESSION

Teaching record











- Student/Tutor
- Booking Class
- My Class Schedule
- My Tutoring Sessions
- Add Tutoring Session
- Teaching Session Record
- Session Records**
- Earnings Dashboard
- Logout

Session Record

DATE	SUBJECT	HOURS	PROOF	STATUS
6/4/2026	Data Structure and Algorithm	2 hrs	View	APPROVED
16/4/2026	Data Structure and Algorithm	2 hrs	View	PENDING

Note : Sessions marked as pending are waiting admin review before being approved for payment

38

Earning dashboard



Home search text



Student/Tutor

Booking Class

My Class Schedule

My Tutoring Sessions

Add Tutoring Session

Teaching Session Record

Session Records

Earnings Dashboard

Logout

Tutor Earnings Dashboard

Total Sessions
2

Total Hours
4

Rate/hours
RM 15

Total Earnings
RM 60

Earnings History

DATE	SUBJECT	HOURS	RATE	AMOUNT	STATUS
6/4/2026	Data Structure and Algorithm	2 hrs	RM 15	RM 30	APPROVED
16/4/2026	Data Structure and Algorithm	2 hrs	RM 15	RM 30	APPROVED
Total Earnings : RM 60				RM 60	

– Admin

Admin Sign in



Student Id/Admin

Password 

☐ Remember me [Forgot Password?](#)

[Log in](#)

[Don't have an account?](#) [Sign up](#)

Verify teaching records



Menu icon

Home icon

Search icon

Bookmarks icon

Notifications icon

Profile icon

Admin

Verify Teaching Record

Approve Payment

Manage User

Logout

Verify Teaching Record

Review and verify teaching session records submitted by tutors.

Submitted Records

No	Tutor Name	Subject	Date	Hours	Proof	Status	Action
1	NURSYUHADAH	DATA STRUCTURE AND ALGORITHM	16-APRIL-2026	1 Hour	VIEW	PENDING	VERIFY REJECT
2	AHMAD ALI	PROGRAMMING	20-APRIL-2026	1 Hour	VIEW	VERIFIED	VERIFIED -

Approve payment









Admin

Verify Teaching Record

Approve Payment

Manage User

Logout

Approve Payment

Review and approve tutor payments.

Submitted Records

No	Tutor Name	Bank Info	Amount (RM)	Hours	Proof	Status	Action
1	NURSYUHADAH	Maybank	30.00	2 Hour	VIEW	PENDING	VERIFY REJECT
2	AHMAD ALI	CIMB Bank	30.00	2 Hour	VIEW	VERIFIED	VERIFY REJECT

Manage user









Admin

Verify Teaching Record

Approve Payment

Manage User

Logout

Manage Users

View and manage user account in the system.

Submitted Records

No	Tutor Name	Email	Role	Status	Action
1	NURSYUHADAH	Syuhadah@gmail.com	Tutor	ACTIVE	EDIT DEACTIVE
2	AHMAD ALI	Ali@gmail.com	Tutor	ACTIVE	EDIT DEACTIVE